

Aurora Cong

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GitHub: <https://github.com/Aurora-1420>

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Education

University of California, San Diego

La Jolla, CA

Bachelor of Science, Data Science

Expected Graduation Date: 2027

Relevant Coursework: Principles of Data Science, Statistical Methods

University of Washington, Seattle

Seattle, WA

Mathematics Major (B.S.), Linguistics Major, GPA: 3.70/4.00

September 2023 - June 2025

Relevant Coursework: Data Structures and Algorithms, Computational Linguistics, Probability, Linear Algebra, Differential Equations, Mathematical Reasoning, Python for Data Analytics, Calculus I/II/III, Introduction to Computer Programming I/II/III

Honors: Dean's List (Autumn 2023, Winter, 2024, Spring 2024, Winter 2025, Spring 2025)

Research Experience

Washington Experimental Mathematics Laboratory

Seattle, WA

Undergraduate Researcher

January 2025 - Present

- Collaborated with Professor Jayadev Athreya of UW-Seattle's Department of Mathematics and a small team of undergraduate researchers on theoretical and computational mathematics research involving the trajectory of mathematical billiards within polygons and the relation of such to musical theory.
- Applied Python/Pygame/Pymunk tools with analytical problem-solving to test hypotheses and visualize and modeling data and mathematical patterns.
- Presented research in weekly meetings.

Linguistics Undergraduate Research Apprentice Program

Seattle, WA

Research Assistant

September 2024 - June 2025

- Designed and ran controlled experiments to study eye-tracking of consonant clusters' pronunciation.

Coding Projects

ELIZALIKE

2025

<https://github.com/Aurora-1420/ELIZA-LIKE->

- Implemented a rule-based natural language processing system inspired by the original ELIZA program.
- Designed pattern-response matching logic to simulate human-like text interactions.
- Gained experience with string manipulation, text processing, and RegEx (Python).

Twitter (X) Trends

2024

<https://github.com/Aurora-1420/Twitter-X--Trends>

- A tool that analyzes tweets (X Posts) to identify trending words and popular posts.
- Users can create and interact with their own posts, with the program providing insights based on trending content.
- Concepts/Tools used: Java, Object Oriented Programming

Skills & Interests

Technical: Advanced in Java and Python, intermediate in HTML/CSS, RegEx, proficient in Praat, Microsoft Suite

Languages: Native fluency in Mandarin Chinese, Intermediate written and conversational French

Interests: Reading, Piano, Cello, Jigsaw Puzzles, Traveling